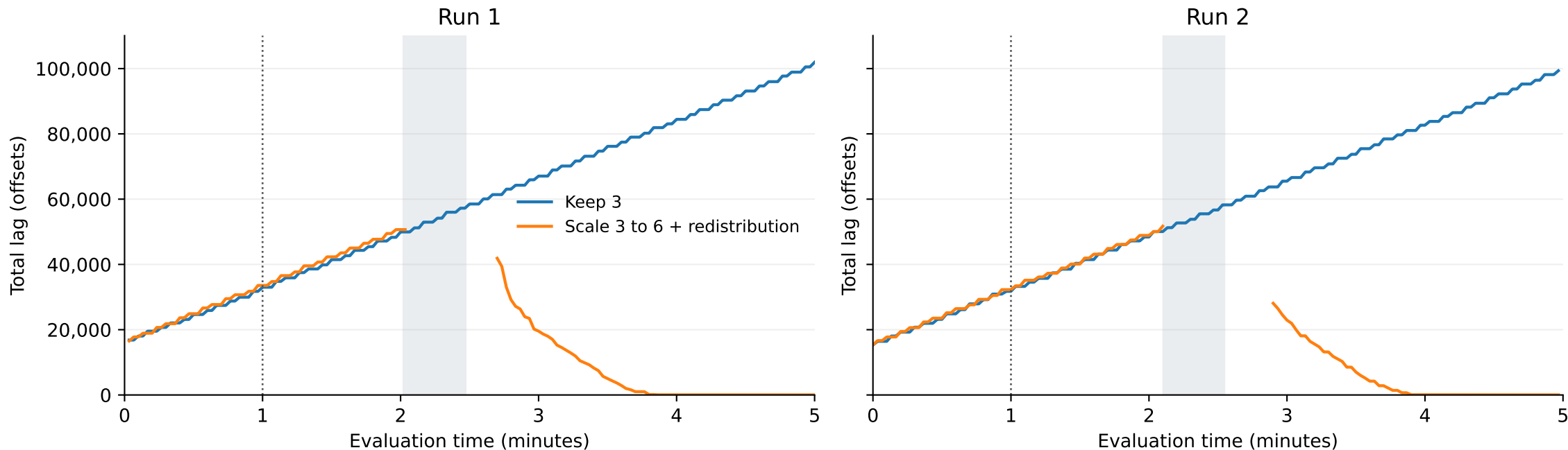


Scaling from concentrated Consumer 2 ownership — 700 messages/s

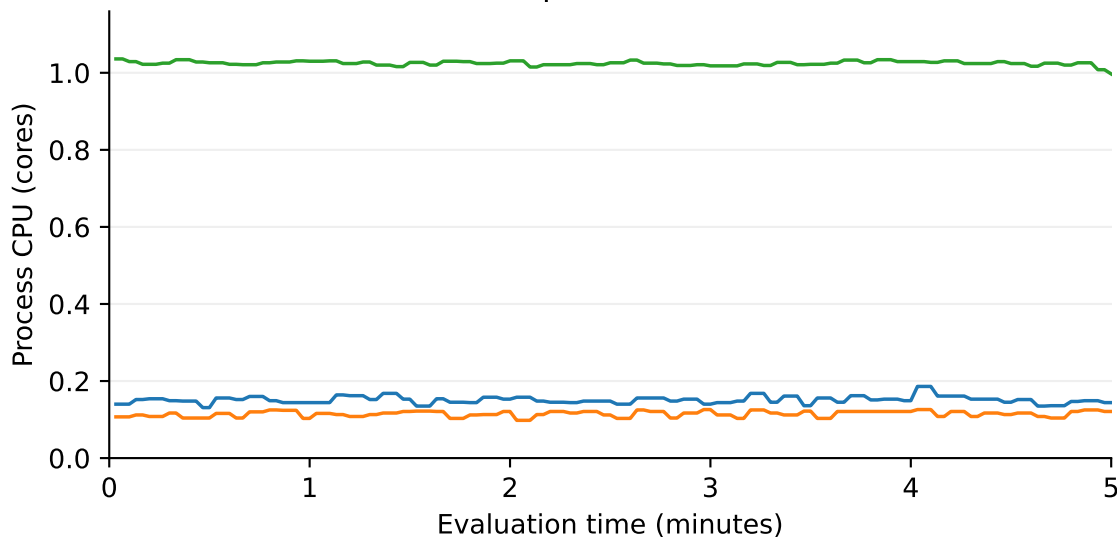


1 min warm-up + 5 min evaluation + 2 min drain. Dotted line: scheduled scale request. Shading: handoff coordination. Evaluation only; gaps remain unavailable. Both runs start with all twelve hot partitions on Consumer 2.

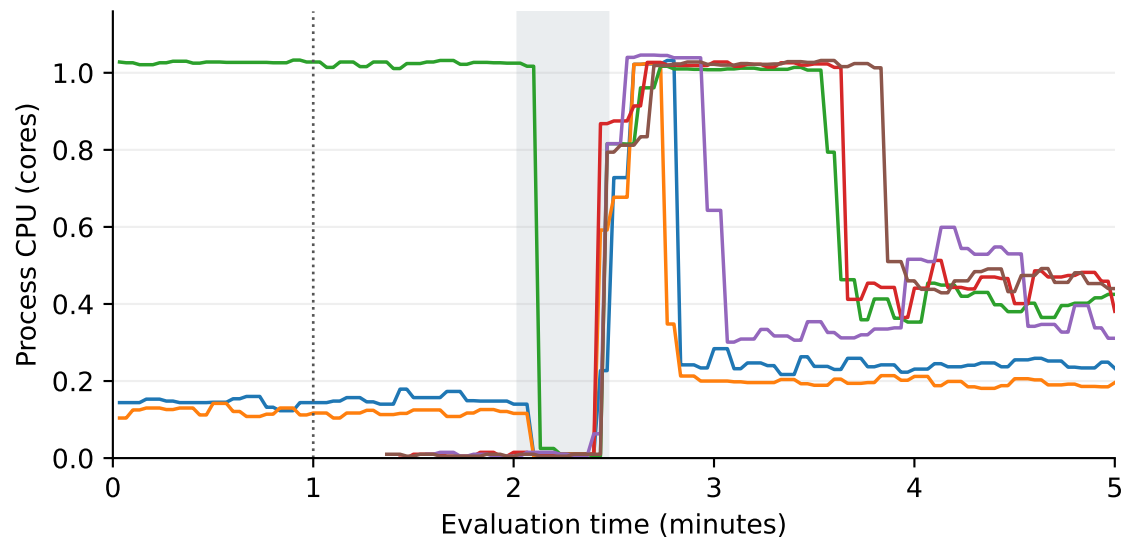
Consumer CPU usage — 700 messages/s

Consumer 0 Consumer 1 Consumer 2 Consumer 3 Consumer 4 Consumer 5

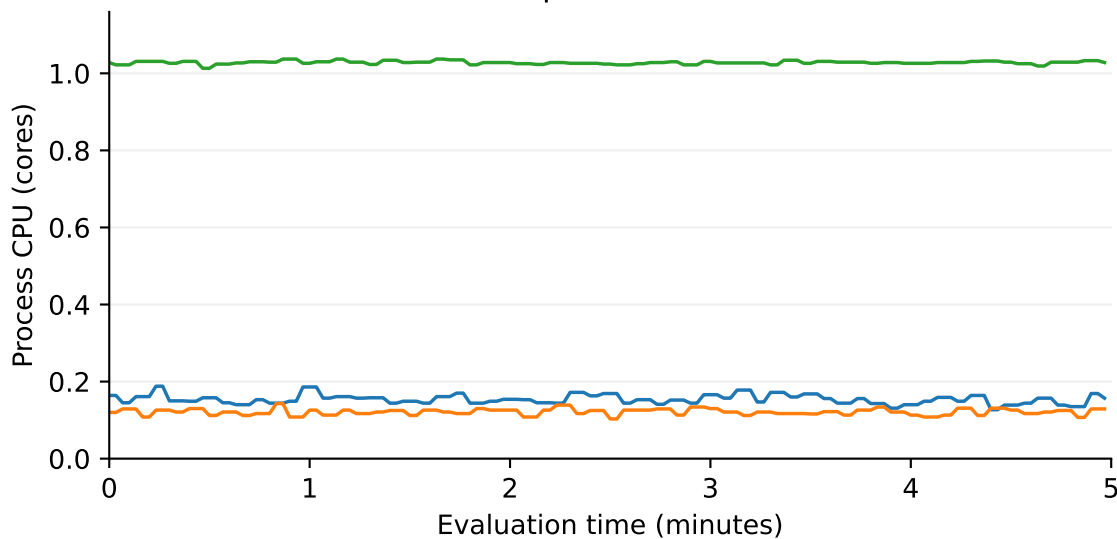
Keep 3 — Run 1



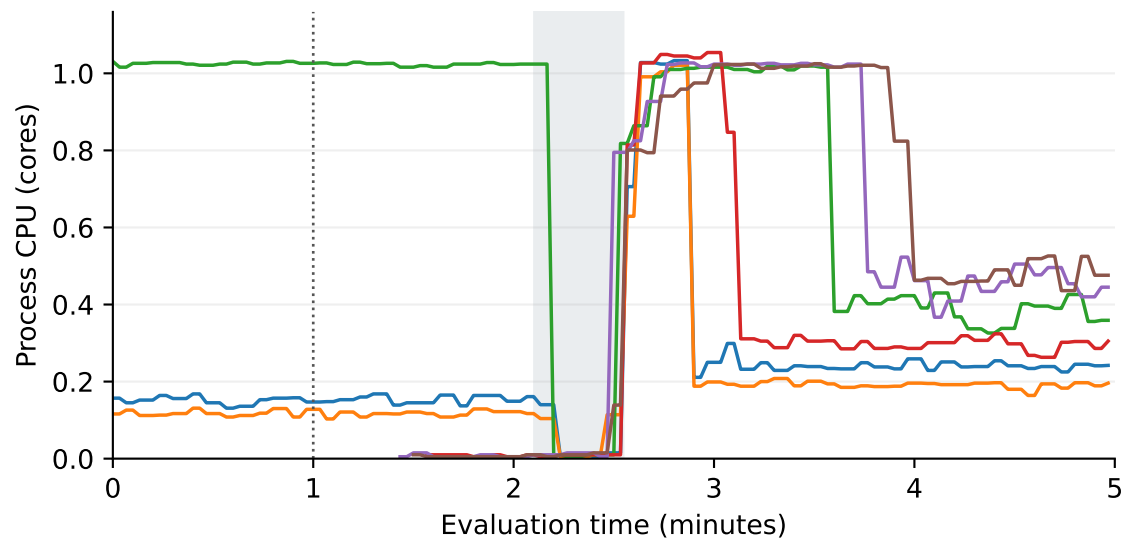
Scale 3 to 6 + redistribution — Run 1



Keep 3 — Run 2



Scale 3 to 6 + redistribution — Run 2

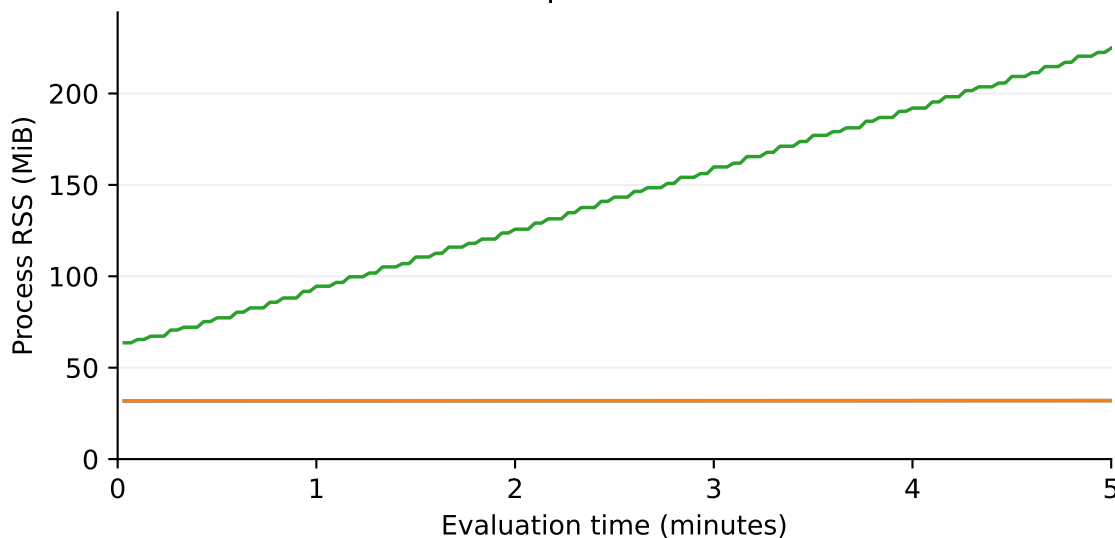


Common scales. Dotted line: scheduled scale request; shading: handoff coordination. Gaps remain unavailable. Process measurements only. New-consumer traces begin when those processes are observed.

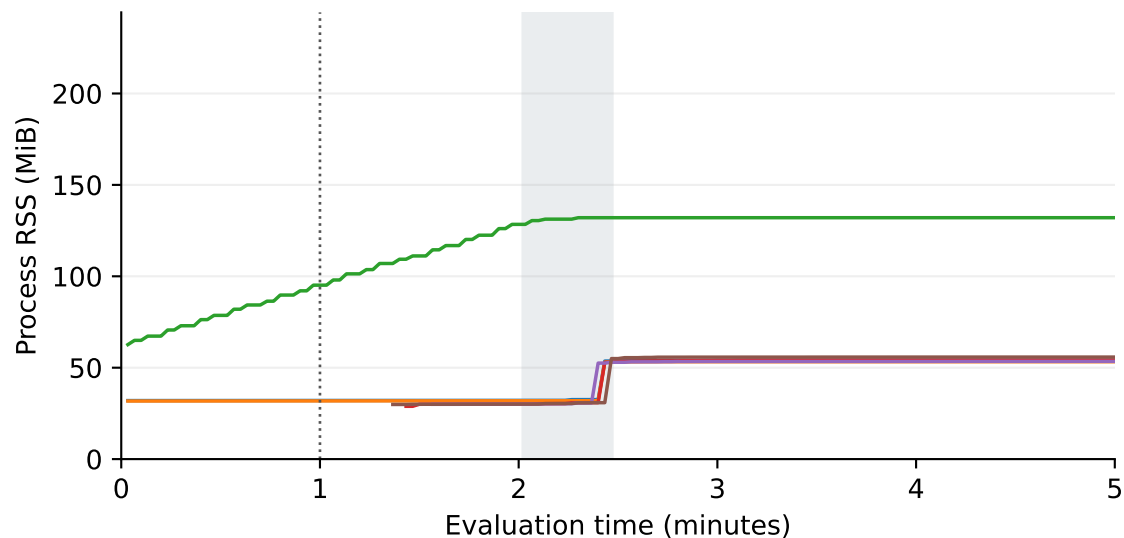
Consumer memory usage — 700 messages/s

Consumer 0 Consumer 1 Consumer 2 Consumer 3 Consumer 4 Consumer 5

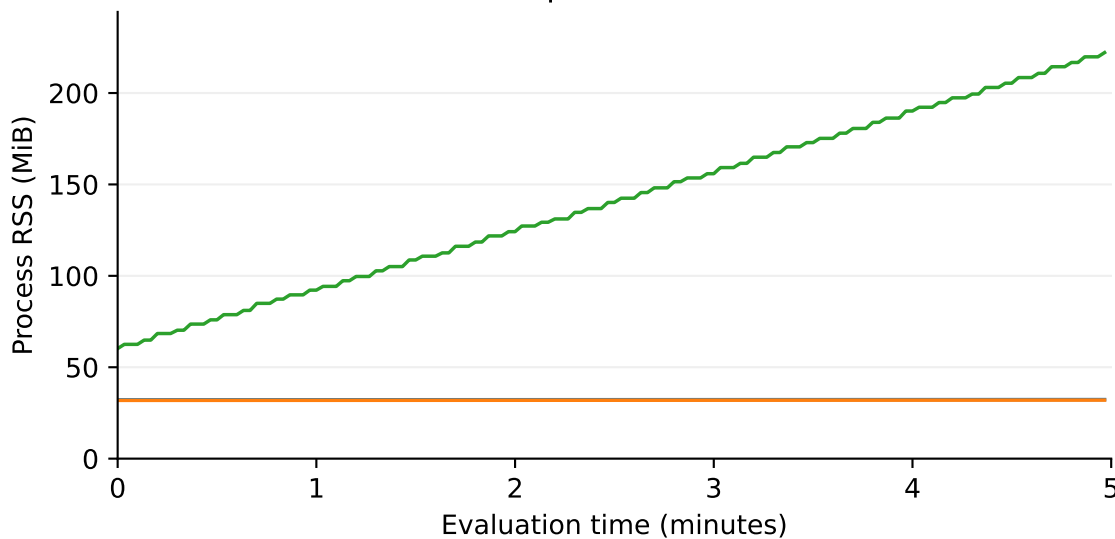
Keep 3 — Run 1



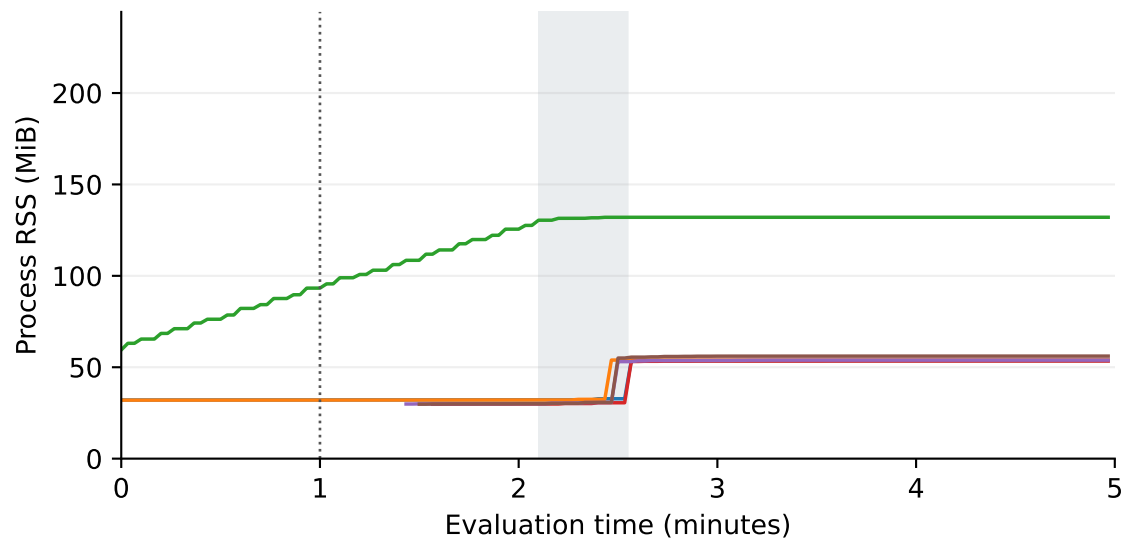
Scale 3 to 6 + redistribution — Run 1



Keep 3 — Run 2



Scale 3 to 6 + redistribution — Run 2

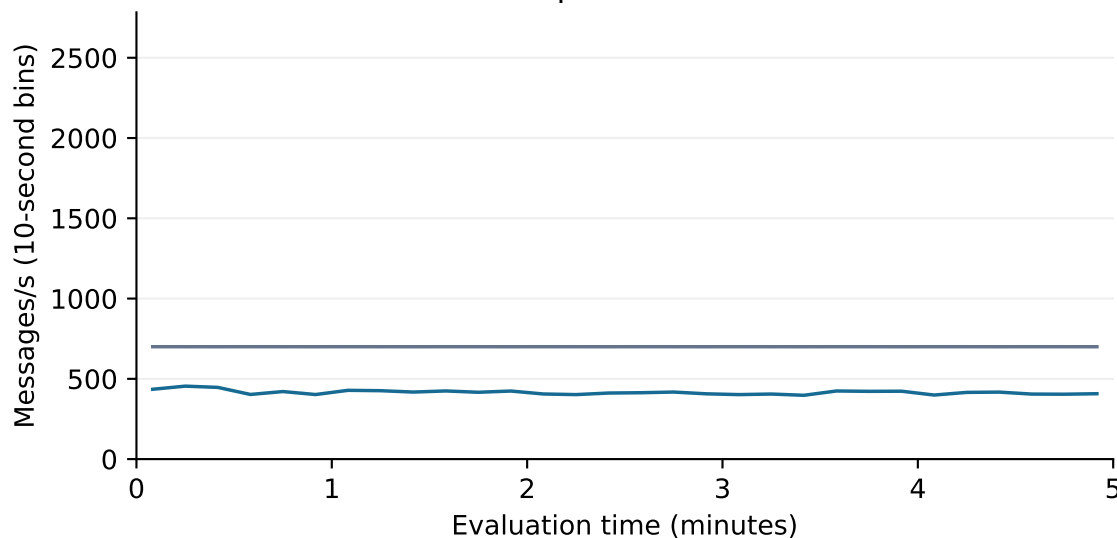


Common scales. Dotted line: scheduled scale request; shading: handoff coordination. Gaps remain unavailable. Process measurements only. New-consumer traces begin when those processes are observed.

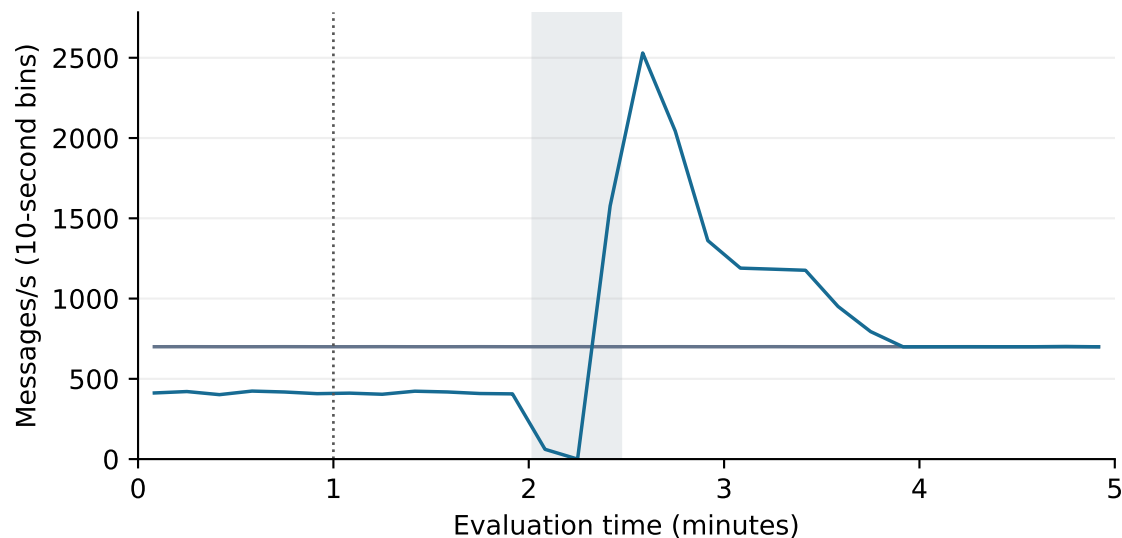
Input and completion throughput — 700 messages/s

Acknowledged input Unique completions

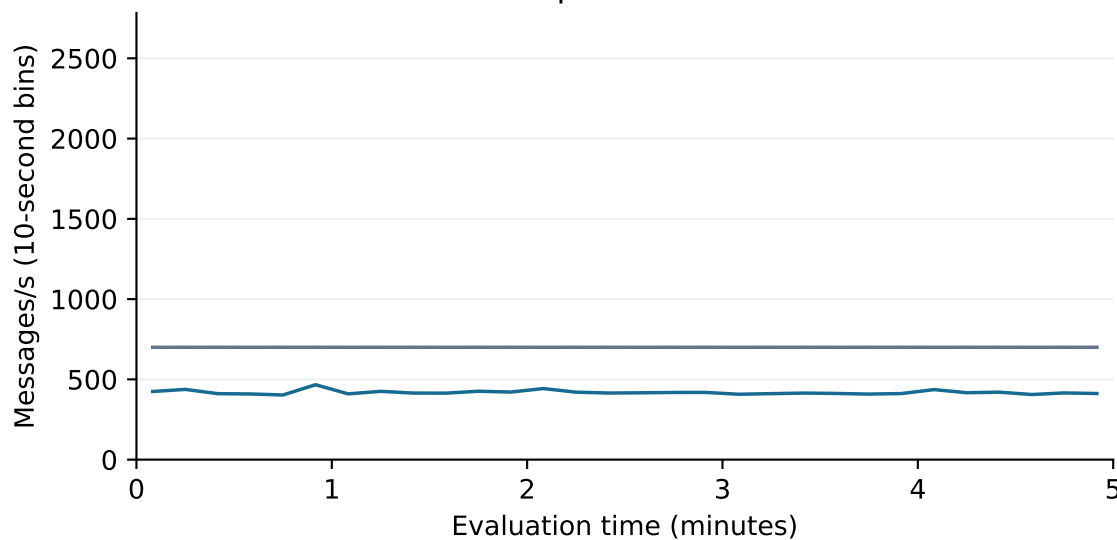
Keep 3 — Run 1



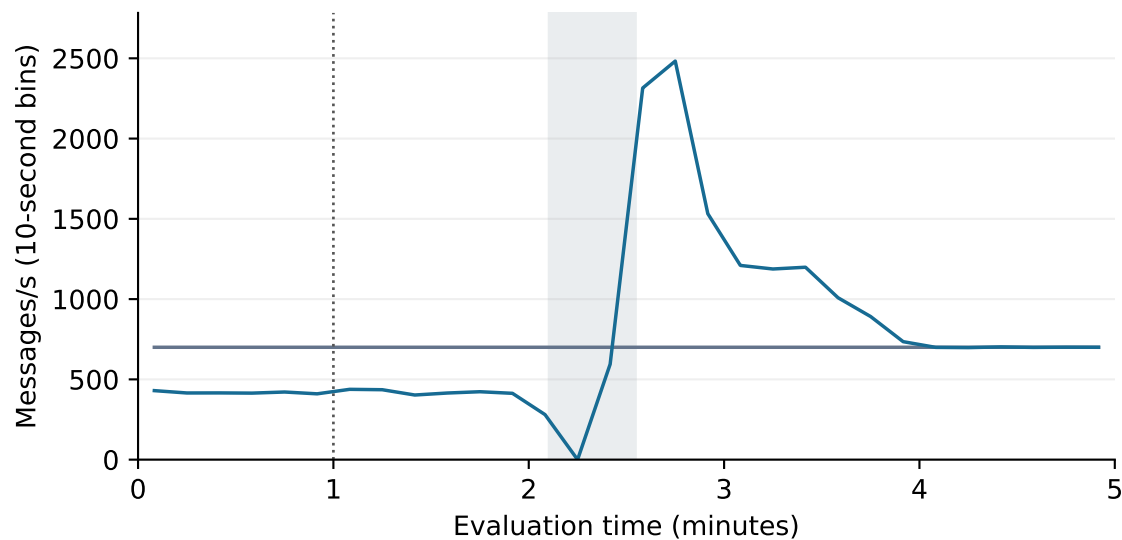
Scale 3 to 6 + redistribution — Run 1



Keep 3 — Run 2



Scale 3 to 6 + redistribution — Run 2

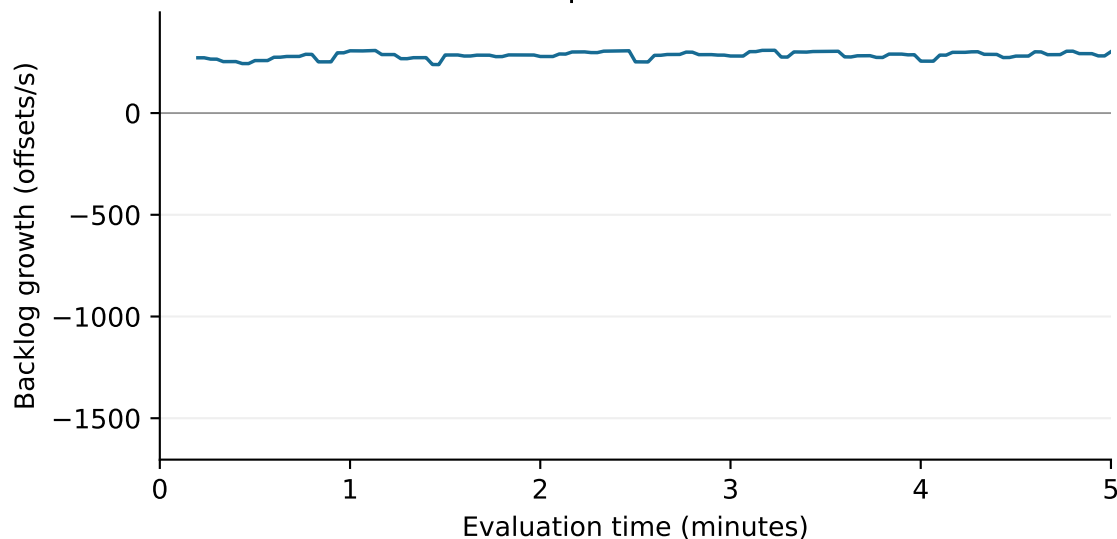


Common scales. Dotted line: scheduled scale request; shading: handoff coordination. Gaps remain unavailable. Completions include warm-up records finishing during evaluation.

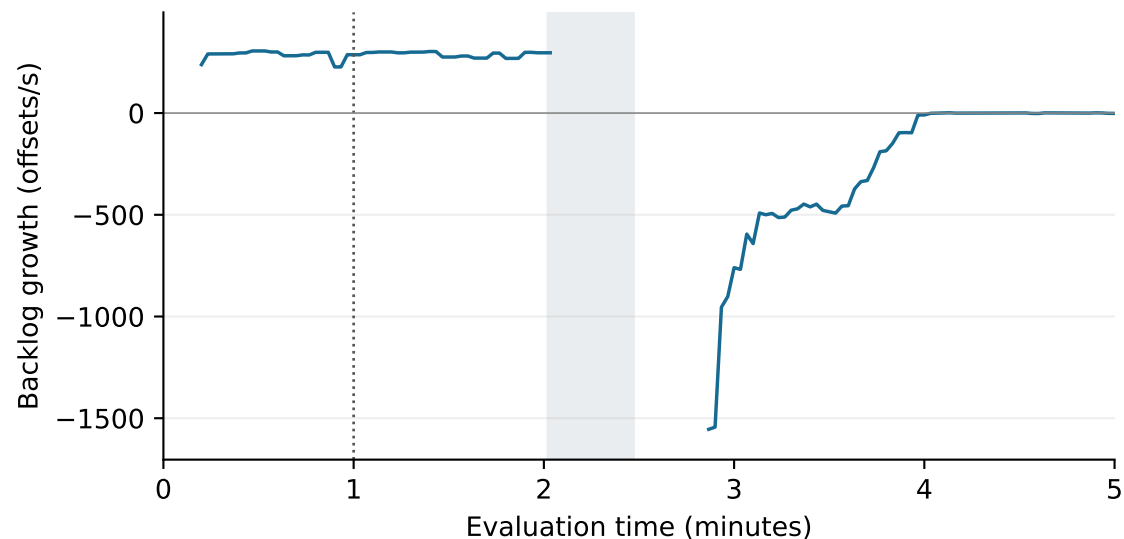
Processing-backlog growth — 700 messages/s

— 10-second growth

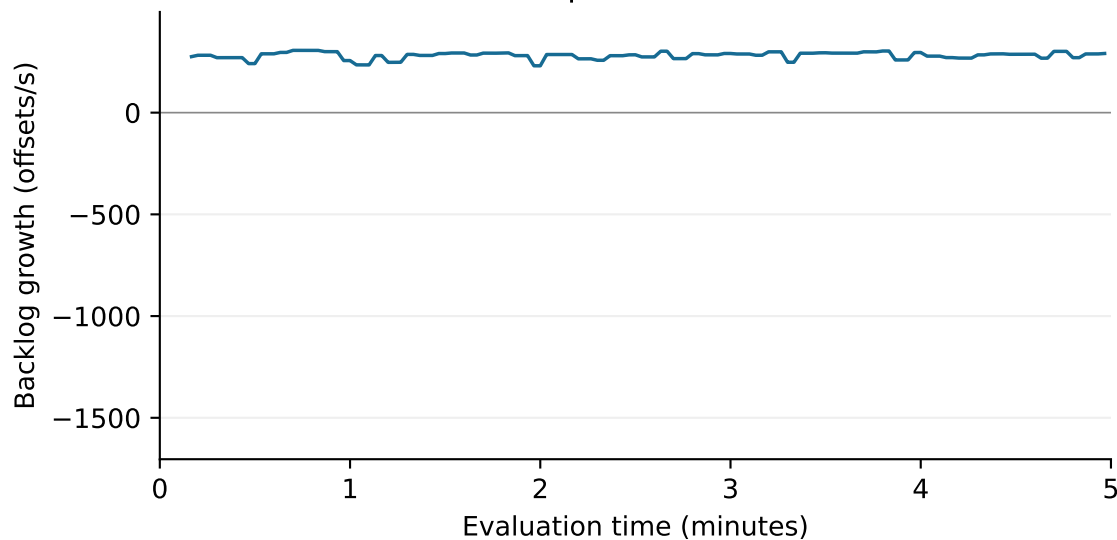
Keep 3 — Run 1



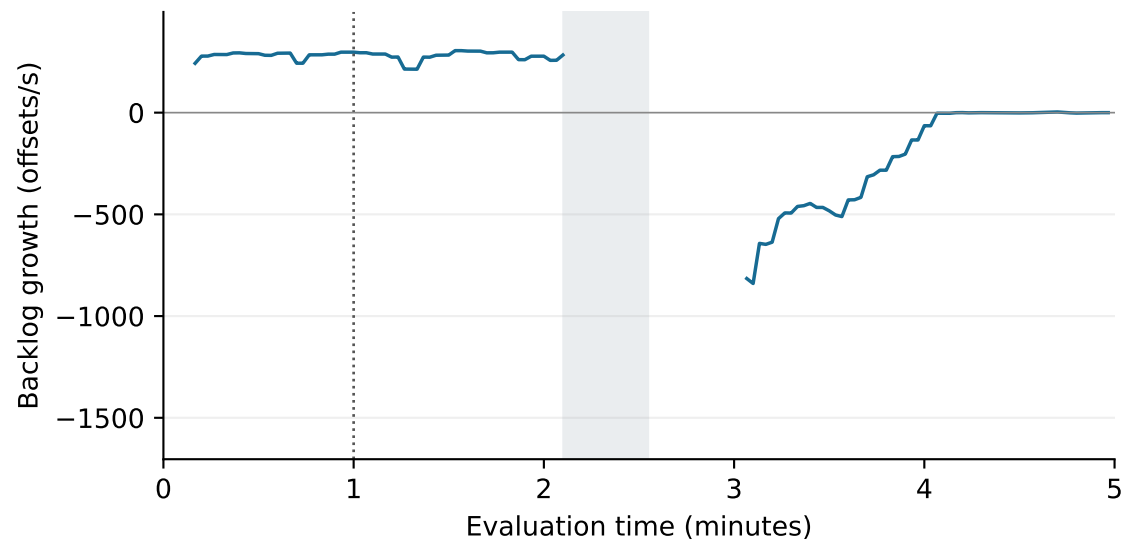
Scale 3 to 6 + redistribution — Run 1



Keep 3 — Run 2



Scale 3 to 6 + redistribution — Run 2

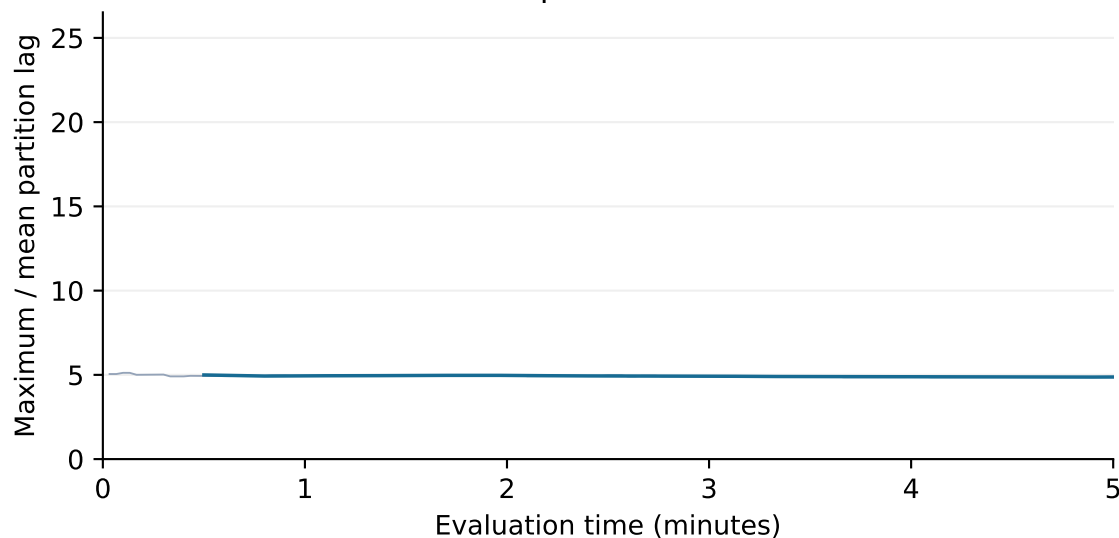


Common scales. Dotted line: scheduled scale request; shading: handoff coordination. Gaps remain unavailable. Growth uses a 10-second window; missing observations are not reconstructed.

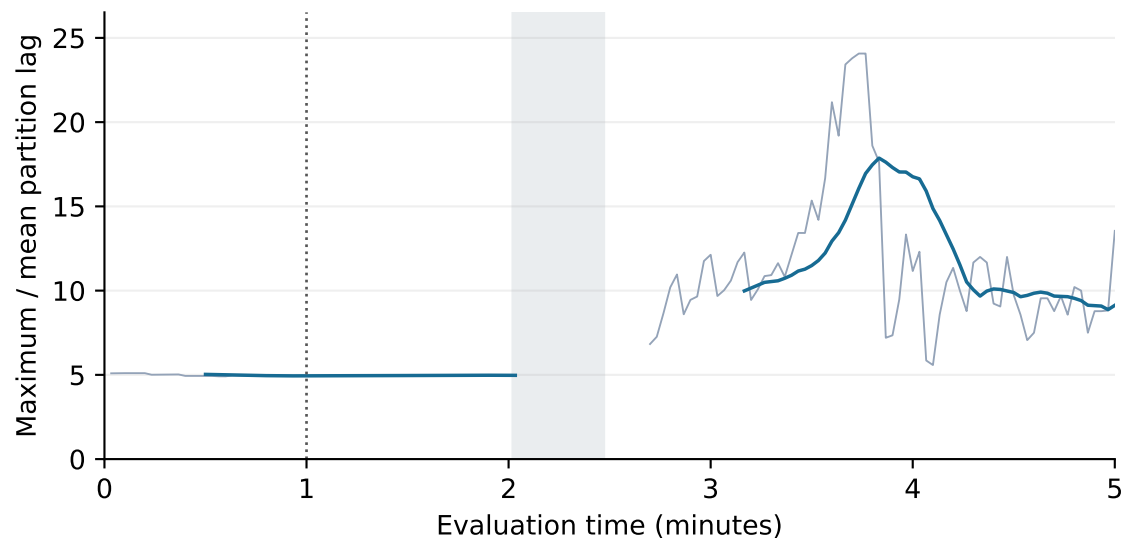
Partition lag skew — 700 messages/s

— Snapshot ratio — 15-sample rolling mean

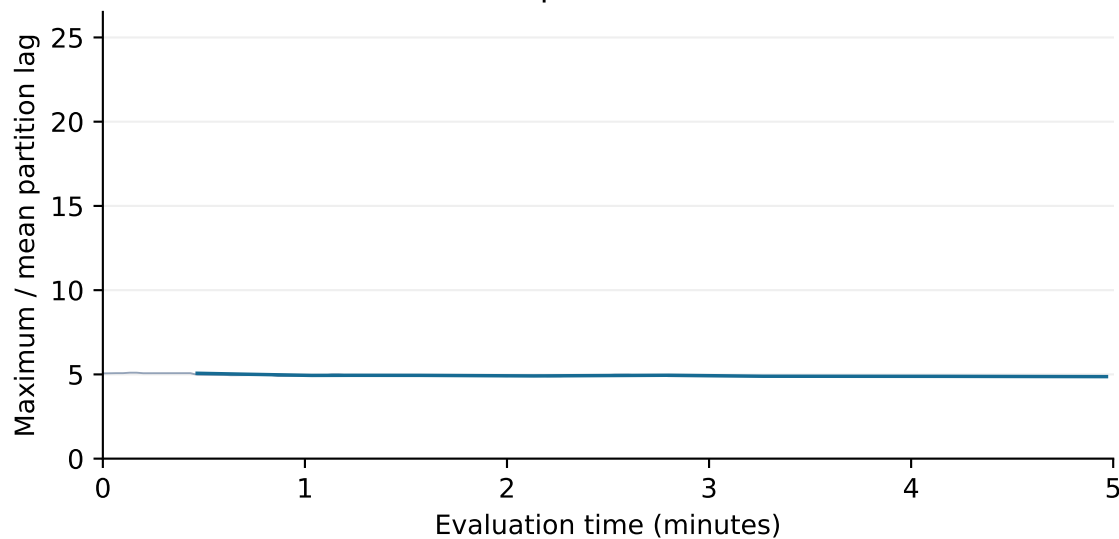
Keep 3 — Run 1



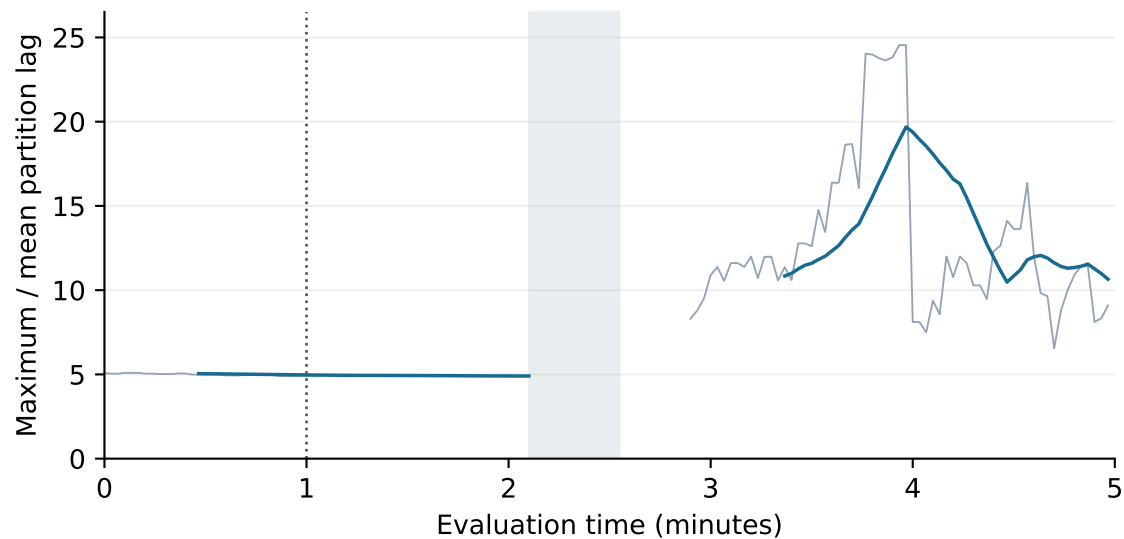
Scale 3 to 6 + redistribution — Run 1



Keep 3 — Run 2



Scale 3 to 6 + redistribution — Run 2

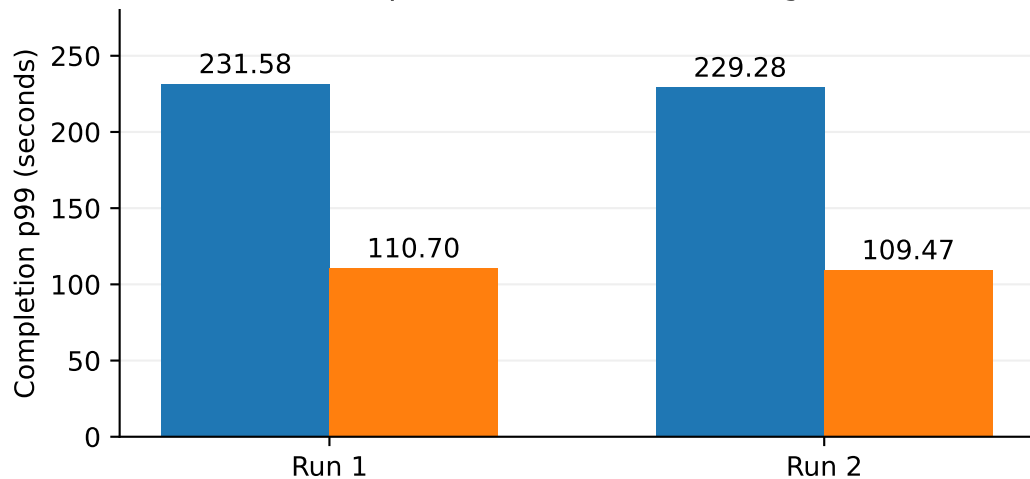


Common scales. Dotted line: scheduled scale request; shading: handoff coordination. Gaps remain unavailable. Interpret relative skew together with absolute lag and growth; a high ratio alone does not imply a large backlog.

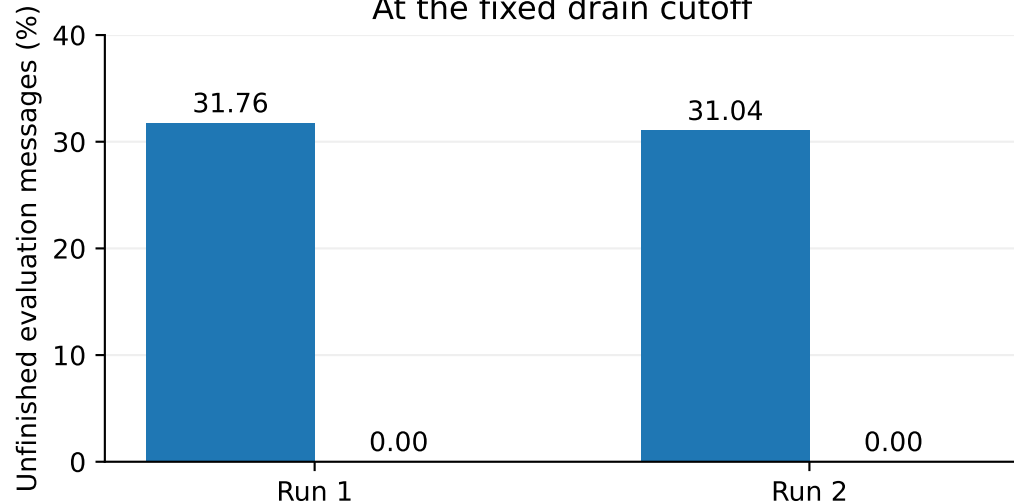
Completion latency and unfinished work — 700 messages/s

Keep 3 Scale to 6 + redistribution

Completed evaluation messages



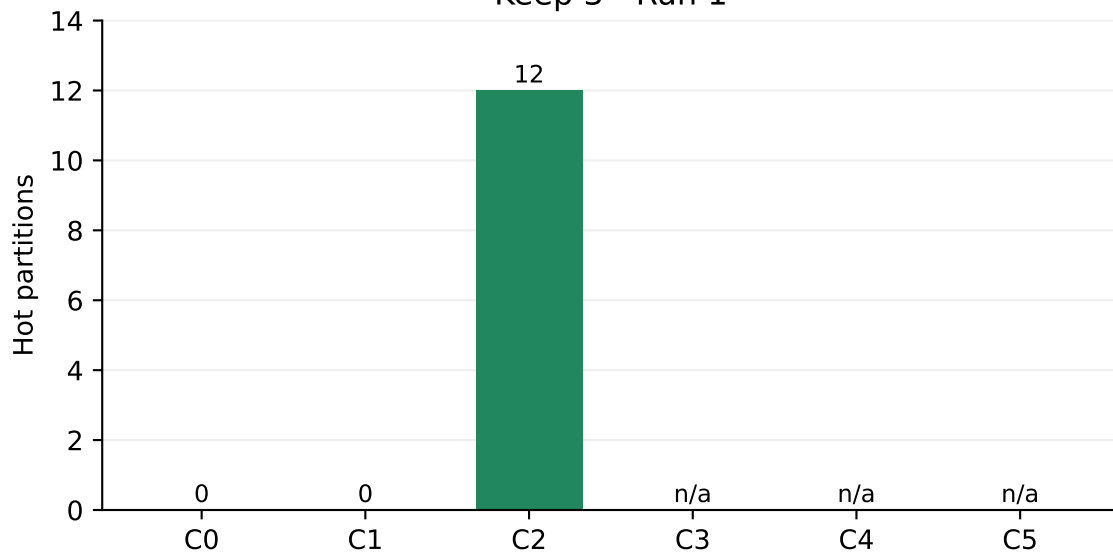
At the fixed drain cutoff



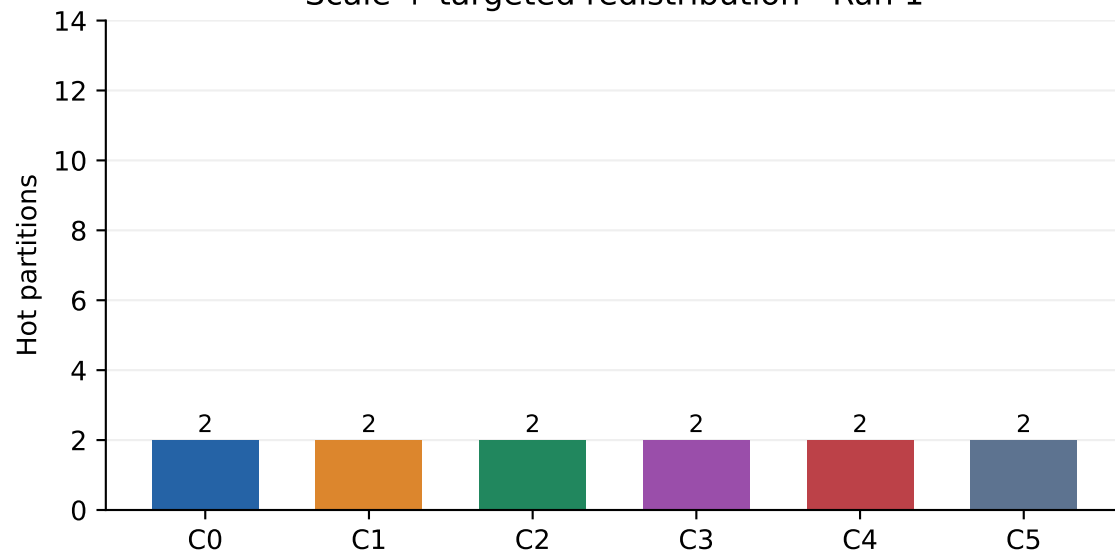
Each trial: 1 min warm-up + 5 min evaluation + 2 min drain; 210,000 evaluation messages.
P99 is a whole-run completed-cohort percentile, not an average of rolling p99 values.

Observed hot-partition ownership at the end of evaluation - 700 messages/s

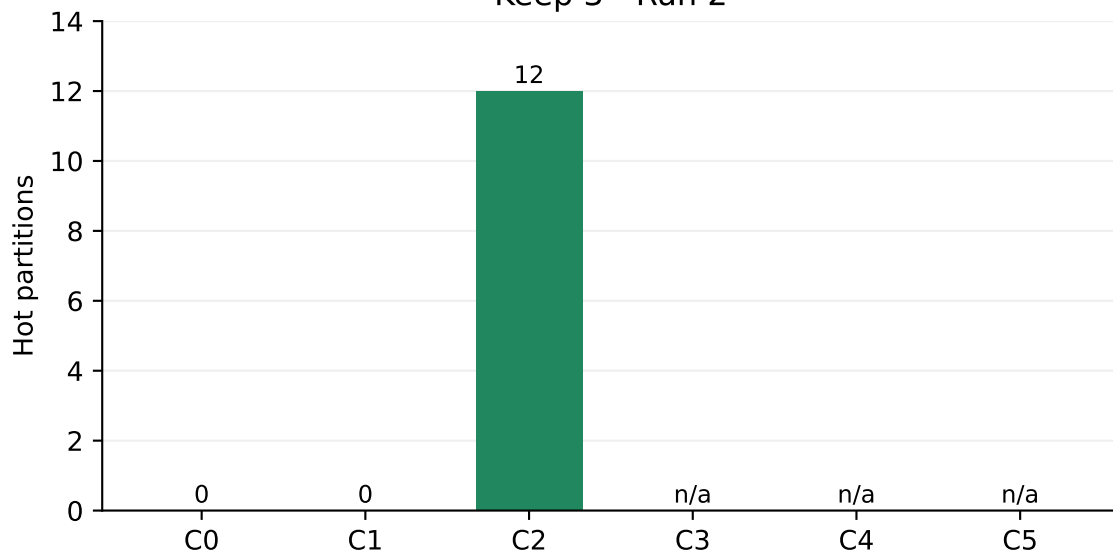
Keep 3 - Run 1



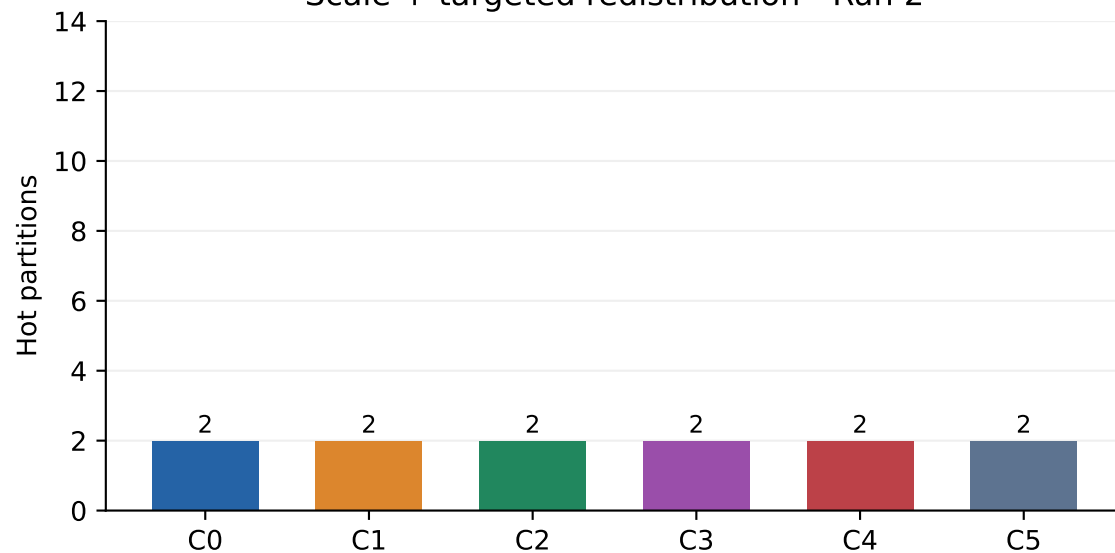
Scale + targeted redistribution - Run 1



Keep 3 - Run 2



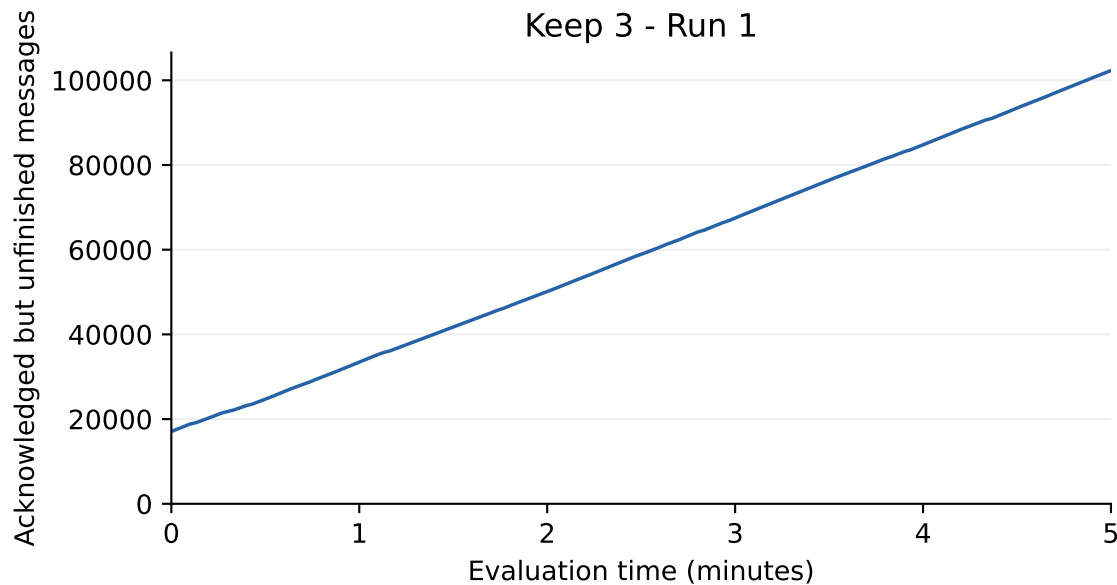
Scale + targeted redistribution - Run 2



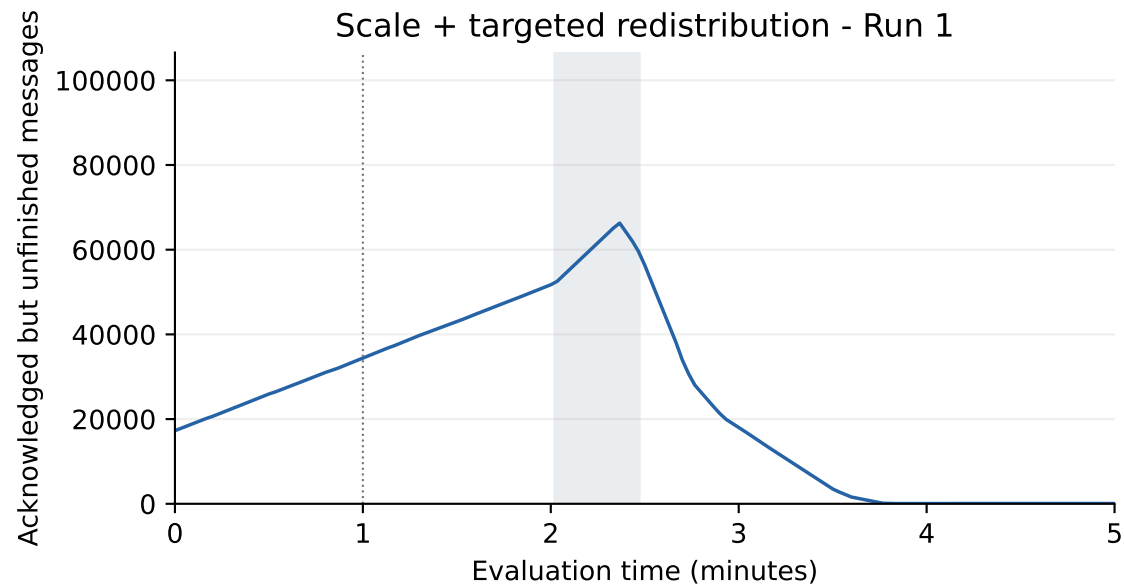
Each trial: 1 min warm-up + 5 min evaluation + 2 min drain. Common scales.
Last complete valid ownership snapshot; n/a means no partition ownership in that snapshot.
C0-C5 identify consumers. Equal partition counts do not imply equal processing load.

Acknowledged but unfinished work - 700 messages/s

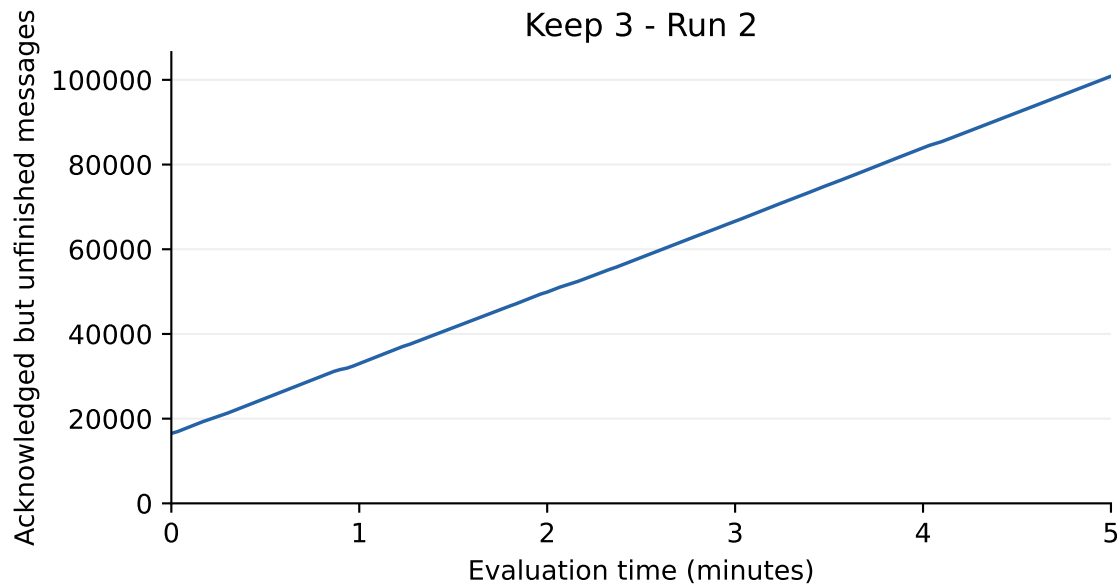
Keep 3 - Run 1



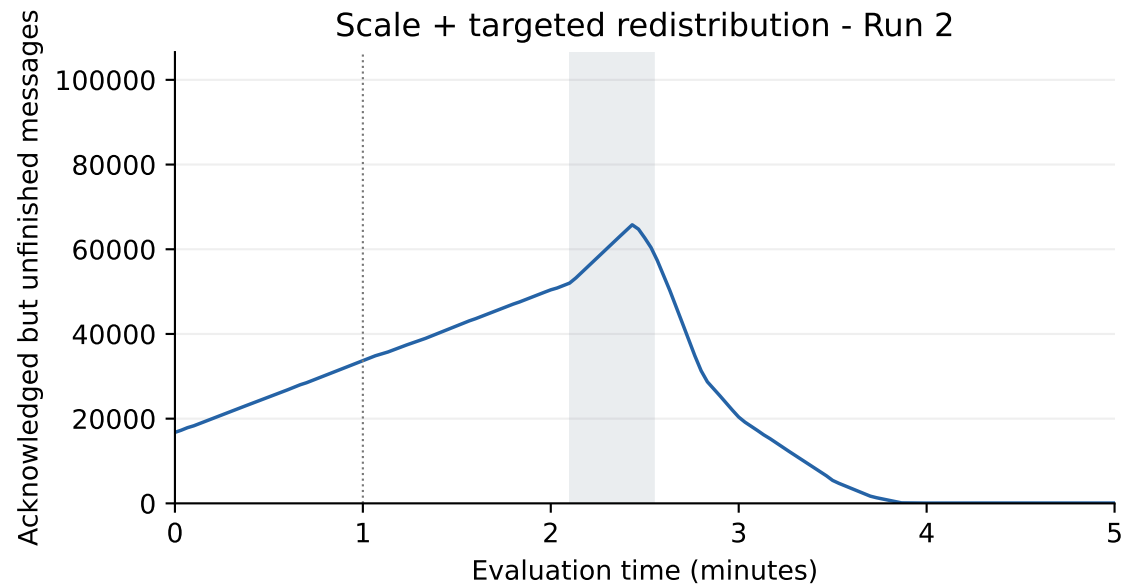
Scale + targeted redistribution - Run 1



Keep 3 - Run 2



Scale + targeted redistribution - Run 2



Each trial: 1 min warm-up + 5 min evaluation + 2 min drain. Common scales.
Distinct ACK/completion events, including warm-up; this is not broker offset lag.
Dotted line: scheduled intervention. Shading: release requested to active verified.